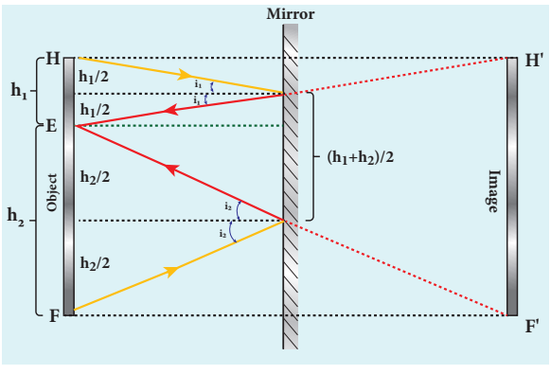


If the distance between his head H and eye E is h_1 and distance between his feet F and eye E is h_2. The person's total height is, h. Here it is, $h = h_1 + h_2$

By the law of reflection, the angle of incidence and angle of reflection are the same for the two extreme reflections. The normals are now the bisectors of the angles between the incident and the reflected rays at the two points. By geometry, the height of the mirror needed is only half of the height of the person. $\frac{h_1 + h_2}{2} = \frac{h}{2}$

Does the height depend on the distance between the person and the mirror?

6.2 SPHERICAL MIRRORS

We shall now study about the reflections that take place in spherical surfaces.

A spherical surface is a part cut from a hollow sphere. Spherical mirrors are generally constructed using glass. One surface of the glass is silvered. The reflection takes place at the other surface which is polished. **If the polished surface of the mirror is convex it is called as *convex mirror.*** These are shown in Figure 6.6.

We shall get familiarised with some of the terminologies pertaining to spherical mirrors.

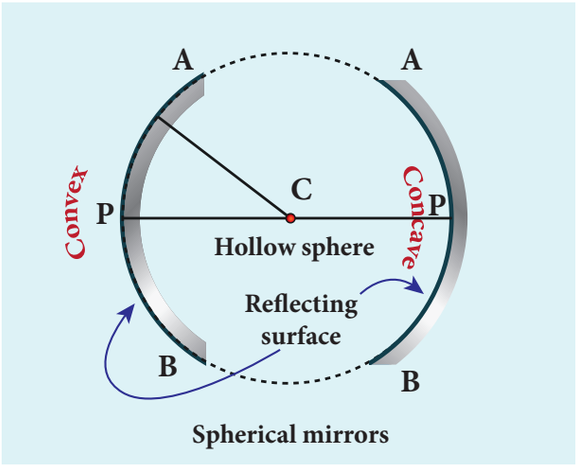


Figure 6.6 Spherical mirrors

Centre of curvature: The centre of the sphere of which the mirror is a part is called the *centre of curvature C* of the mirror.

Radius of curvature: The radius of the sphere of which the spherical mirror is a part is called the *radius of curvature R* of the mirror.

***Pole*:** The middle point on the spherical surface of the mirror (or) the geometrical centre of the mirror is called *pole P* of the mirror (or) optic centre.

Principal axis: The line joining the pole *P* and the centre of curvature *C* is called the *principal axis* of the mirror. The light ray travelling along the principal axis towards the mirror after reflection travels back along the same principal axis. It is also called optical axis

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